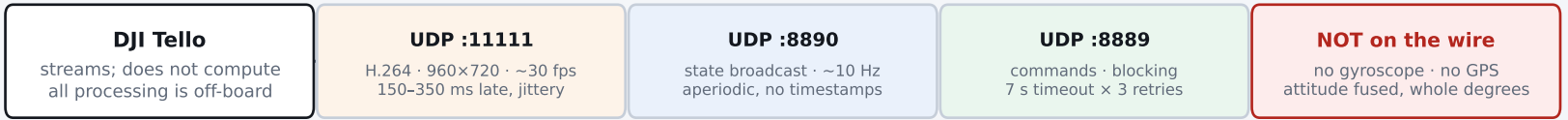


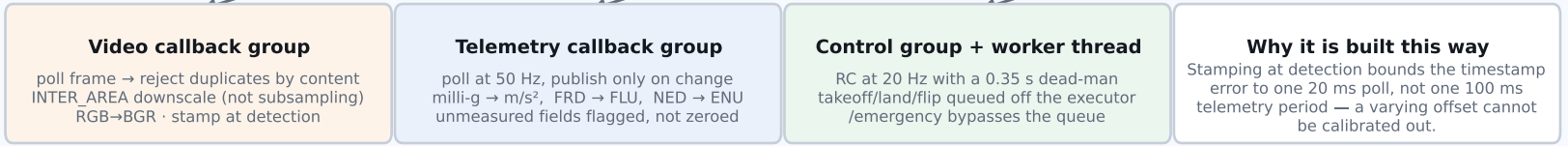
Tello VIO — end-to-end pipeline

from UDP packet to published pose: the rate, the data and the algorithm at every stage

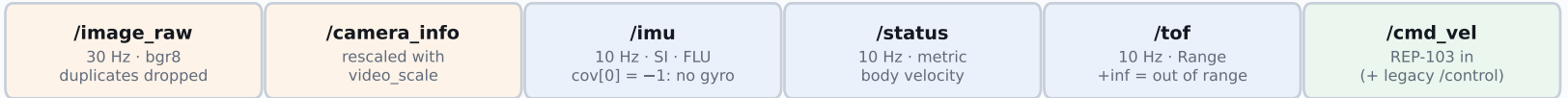
1 • AIRCRAFT



2 • DRIVER • `tello/node.py` • `MultiThreadedExecutor`, 3 callback groups + 1 worker thread — no SDK call blocks a callback

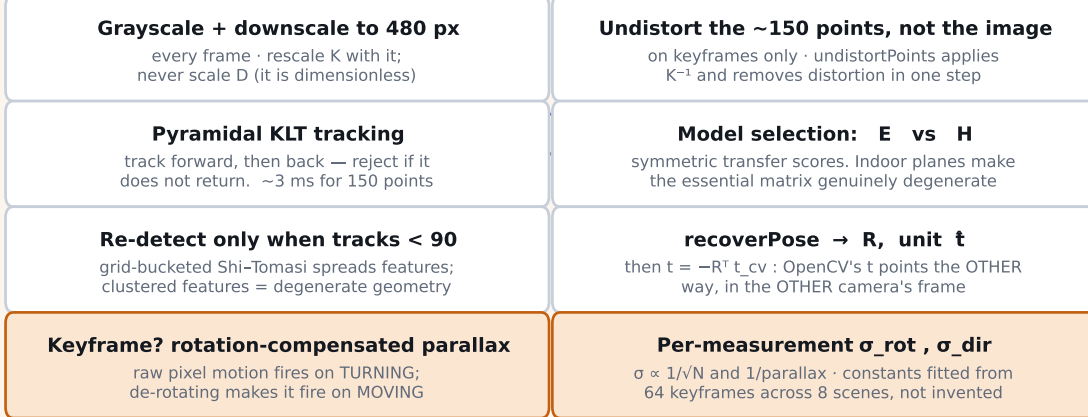


3 • ROS 2 TOPICS • sensor-data QoS: best-effort, depth 1

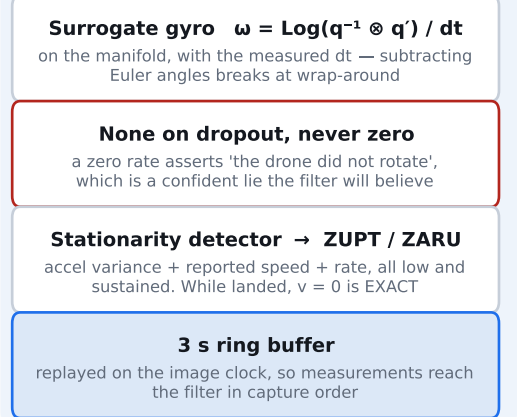


4 • PER-FRAME PROCESSING

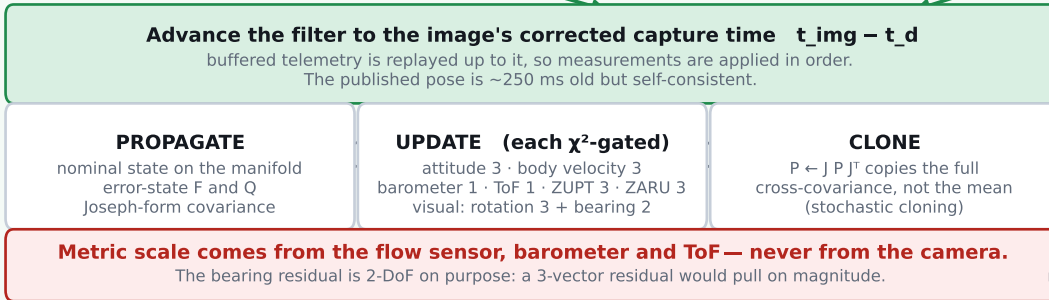
VISUAL FRONT-END • `frontend.py` + `two_view.py`



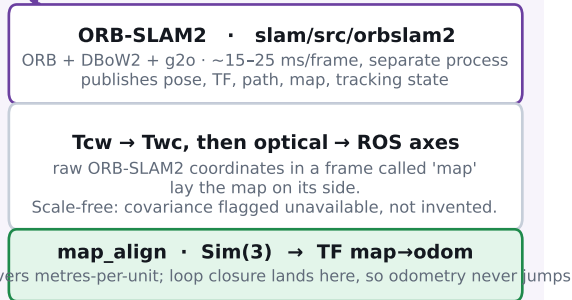
TELEMETRY CONDITIONING • `tello_model.py`



5 • FUSION • `eskf.py` • 22 error states



OPTIONAL BACKEND • loop closure



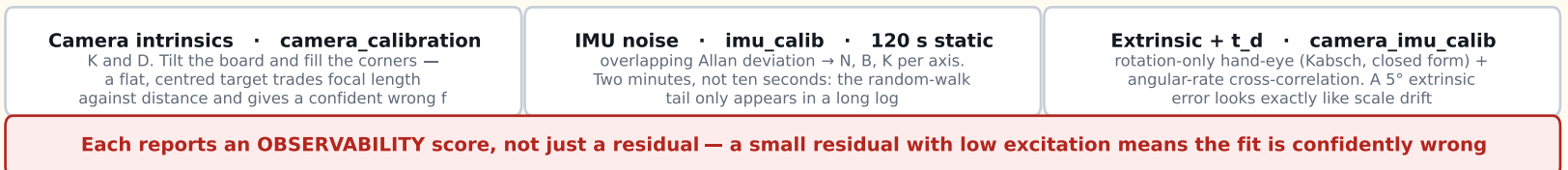
6 • OUTPUTS



TF tree: map —(may jump)—> odom —(continuous)—> base_link —(static, calibrated)—> camera_optical · exactly one node owns each edge

parameters: R_{BC} , t_d , σ_a , σ_ω , K , D

7 • CALIBRATION • offline, once per airframe — these feed PARAMETERS into every stage above



WHAT THIS PIPELINE GUARANTEES — AND WHAT IT DOES NOT

Guarantees

- ✓ Metric output with no GPS: 0.57 m RMS over 40 m flown (1.4 % of path)
- ✓ Roll and pitch bounded and absolute, 1.8° RMS — gravity-referenced
- ✓ Honest covariance: NEES 3.2–3.8 against an ideal 3.0
- ✓ Unmeasured quantities are representable as unmeasured, not as zero

Does not

- ✗ Bound yaw — no magnetometer, ~13° RMS drift over 60 s
- ✗ Survive a dead flow sensor without vision (20× worse)
- ✗ Relocalise in the fast path — KLT carries no descriptors
- ✗ Replace validation on real bags: every figure here is simulated